California Single-Family Home Price Prediction

ML Pipeline using 2025 California Regional Multiple Listing Service (CRMLS) Data

Weighted Stacking Ensemble · LightGBM + XGBoost + Random Forest · $R^2 = 0.90$ · MdAPE = 7.40%

Yanming Kuang · Erdős Institute Data Science Boot Camp Project Presentation · 2025

Introduction

The Challenge

Predict the final closing price of California single-family homes using MLS transaction records — a regression problem where prices range from under $200K to over $10M.

Why It Matters

Home prices are shaped by many factors — location, size, school quality, coastal proximity — in ways that a simple linear model struggles to capture. Tree-based ensemble methods handle these non-linearities naturally.

My Goal

Build a production-ready ML pipeline that agents, lenders, and buyers can use for real-time price estimates with measurable accuracy.

Data Overview

Data Source: California Regional Multiple Listing Service (CRMLS) — 2025 MLS Transaction Records

Note: Top and bottom 0.5% of sale prices removed prior to modeling to prevent unusual transactions from dominating error metrics.

Jan–Nov	Dec	~125K
Training Set	**Test Set**	**Transaction Records**
2025	2025	CRMLS 2025

Key Input Features

- Living Area (sq ft)
- Bedrooms & Bathrooms
- Lot Size
- ZIP Code (prefix-grouped)
- Flooring Type (target encoded)
- High School District (target encoded)
- Distance to Coast

Pipeline Structure

- 01_preprocessing.ipynb → Data cleaning & encoding
- 02_modeling.ipynb → Model training & evaluation
- 03_app.py → Streamlit prediction app
- airflow_dag_pipeline.py → Scheduled automation

Feature Engineering

01

ZIP-Prefix Grouping

ZIP codes grouped by 3-digit prefix to reduce cardinality while preserving geographic signal. Rare ZIPs collapse into regional clusters.

02

Target Encoding

Flooring type and high school district encoded using mean closing price per category. For any unseen category at prediction time, the app falls back to the overall training mean.

03

Distance to Coast

Euclidean distance from each listing to the nearest California coastline point. Captures premium for ocean proximity not captured by ZIP alone.

04

StandardScaler

All numeric features scaled using a StandardScaler fit on the training set only, then applied consistently to the test set and the Streamlit app.

Models Evaluated

Baseline

OLS Regression

Tree-based

Decision Tree

Ensemble

Random Forest

Boosting

XGBoost

Boosting

LightGBM

★ Best Model

Weighted Stacking

LightGBM + XGBoost + RF

Model Results

Best Model: Weighted Stacking Ensemble (LightGBM + XGBoost + Random Forest)

0.90

R^2 Score

Overall

7.40%

MdAPE

Median Absolute % Error

Train: Jan–Nov 2025

Test: December 2025

📊 *Benchmark: Zillow California off-market median error = 6.34%*

Key Takeaway

The model explains 90% of the variance in California single-family home closing prices. The mid-price range ($500K–$1M) — where nearly half of all sales occur — achieves the best MdAPE at 6.32%, while performance degrades at the high end due to data sparsity and luxury-specific features not captured in MLS listings.

Price Band Analysis

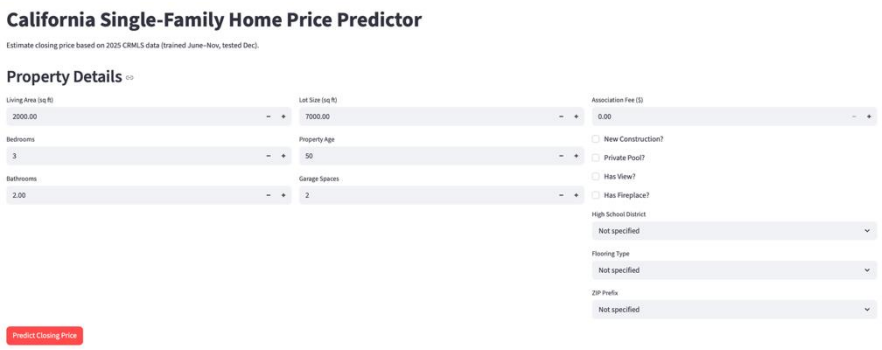
Price Band	R^2	MdAPE	Count
< $500K	0.12	7.99%	1,516
$500K – $1M	0.40	6.32%	4,598
$1M – $2M	0.19	8.58%	2,843
$2M – $5M	0.42	11.64%	921
> $5M	-2.41	16.10%	111

Interactive Streamlit App

How It Works

Demo: [House Price Predictor]

1. User inputs property features (area, beds, baths, ZIP, flooring, lot size)
2. App loads best single model: LightGBM (best_lgbm_model.pkl)
3. Same preprocessing applied: target encoding + StandardScaler
4. Returns estimated closing price in dollars instantly



Automated Pipeline (Apache Airflow)

1

Scheduled FTP Download

DAG checks for new MLS data files. Only downloads files newer than existing local months — avoids redundant re-processing.

2

Preprocessing

Raw CSVs cleaned, imputed, encoded, and scaled into train/test sets using the same logic as the notebooks.

3

Model Retraining

Pipeline retrains the full stacking ensemble on updated data and evaluates on held-out test set.

4

Evaluation & Logging

R^2, MdAPE, and price-band metrics logged. Model artifacts (.pkl files) updated for the Streamlit app.

Next Steps & Limitations

High-End Segment

MLS data only captures basic listing attributes (beds, baths, sq ft). Luxury pricing is driven by fine-grained features — custom finishes, architectural quality, views — that are absent from the dataset.

Geospatial Features

Incorporate walk score, school ratings API, and crime index for richer location signals beyond ZIP and coast distance.

Temporal Trends

Average monthly ZIP-level price changes were tested but did not improve model performance.

Live Retraining

Airflow pipeline ready to retrain monthly as new CRMLS data becomes available via scheduled FTP sync.

$R^2 = 0.90$ · MdAPE = 7.40% · LightGBM + XGBoost + RF Stacking · Production Streamlit App · Airflow Automation